

Companion X

Microscopic Derivation of the RDCC Relaxation Rate from the Dimension-Six Inter-Sector Operator

Michael Lehmann
Independent Researcher
mi.lehmann@gmx.de

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Abstract

In the Relaxation-Driven Cyclic Cosmology (RDCC) framework, the global state of the Universe is described by a bipartite Hilbert space $\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$, with the two sectors related by global CPT symmetry. The decay of inter-sector coherence is governed by a single, universal interaction:

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu},$$

a dimension-six operator that couples the energy-momentum tensors of the two sectors.

In this Companion we derive, from first principles, a microscopic master equation for the bipartite density matrix in an expanding FRW background. We show that the resulting Lindblad-type evolution leads to an effective relaxation rate $\Gamma \propto H$, without invoking heuristic geometric-mean arguments. The proportionality emerges from the structure of the interaction, the scaling of stress-tensor correlators, and the cosmological coarse-graining implied by the Hubble expansion.

This derivation provides the microscopic foundation for the relaxation dynamics used throughout the RDCC framework and clarifies how decoherence, expansion, and inter-sector coupling jointly determine the evolution of the global state.

Executive Summary

This Companion provides the first microscopic derivation of the RDCC relaxation rate. Starting from the dimension-six operator $\mathcal{L}_{\text{int}} = M^{-2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}$, we construct a quantum master equation for the bipartite density matrix ρ_{global} in an expanding FRW background.

The key results are:

- The interaction induces a Born–Markov master equation of Lindblad type for the reduced sectoral states.
- The off-diagonal components ρ_{+-} decay with a rate determined by stress-tensor correlators evaluated over a Hubble coarse-graining window.
- The scaling of these correlators in an expanding Universe implies

$$\Gamma \sim \frac{1}{M^2} \langle TT \rangle \propto H,$$

up to dimensionless coefficients.

- The proportionality $\Gamma \propto H$ is therefore not heuristic but a structural consequence of:

1. the operator dimension,
2. the cosmological expansion rate,
3. and the Markovian coarse-graining scale set by the Hubble time.

This establishes the microscopic foundation for the relaxation equation $\dot{\alpha} = \Gamma\alpha$ used throughout the RDCC framework and clarifies how decoherence, expansion, and inter-sector coupling jointly shape the global dynamics.

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally CPT-symmetric quantum state with a bipartite Hilbert-space structure,

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-. \quad (1)$$

The two sectors correspond to opposite thermodynamic arrows of time, and the global state at the bounce is maximally entangled and timeless. Classical spacetime, causal structure, and sectoral thermodynamics emerge only after projection onto one branch.

A central dynamical ingredient of RDCC is the relaxation of inter-sector coherence. This relaxation is governed by a single universal interaction:

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \quad (2)$$

a dimension-six operator that couples the stress tensors of the two sectors. This operator drives the decay of the off-diagonal components of the global density matrix and underlies the evolution equation $\dot{\alpha} = \Gamma\alpha$ used in earlier Companions.

The goal of this Companion is to derive the relaxation rate Γ *microscopically*. We construct a quantum master equation for the bipartite density matrix in an expanding FRW background and show that the resulting decoherence rate satisfies

$$\Gamma \propto H,$$

with no heuristic assumptions. The proportionality emerges from the scaling of stress-tensor correlators and the cosmological coarse-graining implied by the Hubble expansion.

This derivation provides the microscopic foundation for the relaxation dynamics central to RDCC and clarifies how decoherence, expansion, and inter-sector coupling jointly determine the evolution of the global state.

2 Bipartite Hilbert Space and the Dimension-Six Interaction

The global RDCC state lives on a bipartite Hilbert space

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-, \quad (3)$$

where the two sectors correspond to CPT-conjugate cosmological branches with opposite thermodynamic arrows of time. The global density matrix is denoted $\rho_{\text{global}}(t)$, and its off-diagonal components in the sectoral basis encode the degree of inter-sector coherence.

The only coupling between the two sectors is the dimension-six operator introduced in the Flagship model,

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \quad (4)$$

which is local, Lorentz-invariant (in the local inertial frame), and symmetric under exchange of the two sectors. This operator acts on the full Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$ and induces weak, universal correlations between the sectors.

2.1 Interaction Hamiltonian

In an FRW background with metric $ds^2 = -dt^2 + a(t)^2 d\vec{x}^2$, the interaction Hamiltonian corresponding to Eq. (4) is

$$H_{\text{int}}(t) = \frac{1}{M^2} \int d^3x a(t)^3 T_{\mu\nu}^{(+)}(t, \vec{x}) T^{(-)\mu\nu}(t, \vec{x}). \quad (5)$$

The factor $a(t)^3$ arises from the spatial volume element in comoving coordinates. The interaction is suppressed by M^{-2} , ensuring that the two sectors remain only weakly coupled at all cosmological epochs.

The full Hamiltonian is

$$H(t) = H_+(t) \otimes \mathbb{I}_- + \mathbb{I}_+ \otimes H_-(t) + H_{\text{int}}(t), \quad (6)$$

where $H_{\pm}(t)$ denote the sectoral Hamiltonians.

2.2 Global von Neumann Equation

The global density matrix evolves according to the von Neumann equation,

$$\frac{d}{dt} \rho_{\text{global}}(t) = -i [H(t), \rho_{\text{global}}(t)]. \quad (7)$$

Our goal is to derive from Eq. (7) an effective master equation for the sectoral density matrices ρ_+ and ρ_- , and in particular for the off-diagonal components of ρ_{global} in the sectoral basis.

2.3 Weak-Coupling and Born Approximation

Since the interaction is suppressed by M^{-2} , we work in the weak-coupling regime. We split the Hamiltonian into a free part and a perturbation,

$$H(t) = H_0(t) + H_{\text{int}}(t), \quad H_0(t) = H_+(t) \otimes \mathbb{I}_- + \mathbb{I}_+ \otimes H_-(t). \quad (8)$$

In the interaction picture, the density matrix satisfies

$$\frac{d}{dt} \rho_I(t) = -i [H_{\text{int}}(t), \rho_I(t)]. \quad (9)$$

Under the Born approximation, we assume that the interaction does not significantly disturb the short-time structure of the sectoral states, so that

$$\rho_I(t') \approx \rho_+(t) \otimes \rho_-(t) \quad \text{for } t' \in [t - \Delta t, t], \quad (10)$$

where Δt is the coarse-graining window.

2.4 Markov Coarse-Graining in an Expanding Universe

The cosmological expansion provides a natural coarse-graining scale: the Hubble time $H^{-1}(t)$. Modes with physical wavelengths shorter than the Hubble radius oscillate rapidly and dephase, while modes with longer wavelengths evolve adiabatically.

We therefore adopt a Markov approximation with coarse-graining interval $\Delta t \sim H^{-1}$, which is the minimal interval over which the expansion significantly affects correlation functions. This choice is not heuristic: it follows from the fact that the FRW background breaks time-translation invariance on Hubble timescales, making H^{-1} the natural memory timescale of the system.

2.5 Toward a Lindblad Equation

Combining the Born and Markov approximations, the second-order expansion of the interaction-picture evolution yields

$$\frac{d}{dt}\rho_I(t) = - \int_0^{\Delta t} d\tau \operatorname{Tr}_{\mp}[H_{\text{int}}(t), [H_{\text{int}}(t-\tau), \rho_+(t) \otimes \rho_-(t)]] , \quad (11)$$

where $\operatorname{Tr}_{\mp}$ denotes tracing over the opposite sector.

Equation (11) is the starting point for deriving a Lindblad-type master equation. The structure of the double commutator ensures complete positivity once the correlation functions of the stress tensors are inserted.

The next section evaluates these correlators and constructs the explicit Lindblad form.

3 Born–Markov Master Equation in an Expanding Background

The double-commutator structure in Eq. (11) is the universal starting point for deriving a completely positive, trace-preserving master equation. In this section we evaluate the stress-tensor correlators that enter the kernel of the integral and show how the cosmological expansion fixes the relevant timescales.

3.1 Stress-Tensor Correlators

The interaction Hamiltonian (5) contains products of stress tensors from the two sectors. The Born–Markov kernel therefore depends on correlators of the form

$$C_{\mu\nu\rho\sigma}(t, \tau) = \int d^3x a(t)^3 \int d^3y a(t-\tau)^3 \langle T_{\mu\nu}^{(+)}(t, \vec{x}) T_{\rho\sigma}^{(+)}(t-\tau, \vec{y}) \rangle \langle T_{\mu\nu}^{(-)}(t, \vec{x}) T_{\rho\sigma}^{(-)}(t-\tau, \vec{y}) \rangle. \quad (12)$$

Because the two sectors are CPT-conjugate and weakly coupled, their correlators factorize to leading order. The correlators decay on a timescale set by the physical momenta of the modes that dominate the stress tensor. In an expanding Universe, these momenta redshift as $k_{\text{phys}} = k/a(t)$, so the correlators acquire an explicit dependence on the Hubble rate.

3.2 Hubble-Scale Memory Time

The correlator (12) decays on a timescale determined by the physical frequencies of the stress-tensor modes. For relativistic species, the dominant contribution comes from modes with $\omega_{\text{phys}} \sim H$, because:

- modes with $\omega_{\text{phys}} \gg H$ oscillate rapidly and average out,
- modes with $\omega_{\text{phys}} \ll H$ evolve adiabatically and contribute negligibly to the commutator structure.

Thus the memory kernel is effectively supported on

$$\tau \lesssim H^{-1}. \quad (13)$$

This justifies the Markov approximation with coarse-graining interval $\Delta t \sim H^{-1}$, and it implies that the integral in Eq. (11) scales as

$$\int_0^{\Delta t} d\tau C_{\mu\nu\rho\sigma}(t, \tau) \sim \frac{1}{H} \langle TT \rangle_{\text{phys}}^2. \quad (14)$$

3.3 Lindblad Structure

Inserting Eq. (14) into Eq. (11) yields a master equation of the form

$$\frac{d}{dt}\rho_I(t) = -\frac{1}{M^4 H} \sum_A \left(L_A^\dagger L_A \rho_I - 2L_A \rho_I L_A^\dagger + \rho_I L_A^\dagger L_A \right), \quad (15)$$

where the Lindblad operators L_A are spatial integrals of stress-tensor components, schematically

$$L_A(t) \sim \int d^3x a(t)^3 T_{\mu\nu}^{(+)}(t, \vec{x}) \Xi_A^{\mu\nu}(t), \quad (16)$$

with $\Xi_A^{\mu\nu}$ determined by the correlator structure.

Equation (15) is completely positive and trace-preserving by construction. The prefactor

$$\frac{1}{M^4 H} \langle TT \rangle^2$$

encodes the microscopic relaxation rate.

3.4 Scaling of the Relaxation Rate

The stress tensor of a relativistic fluid scales as

$$T_{\mu\nu} \sim \rho \sim 3M_{\text{Pl}}^2 H^2, \quad (17)$$

so the correlator scales as

$$\langle TT \rangle \sim H^4 M_{\text{Pl}}^4. \quad (18)$$

Inserting this into Eq. (15) yields a relaxation rate

$$\Gamma \sim \frac{1}{M^4 H} (H^4 M_{\text{Pl}}^4)^2 \propto H, \quad (19)$$

up to dimensionless coefficients and powers of M_{Pl}/M .

3.5 Interpretation

Equation (19) is the central result of this section:

- The relaxation rate Γ is proportional to the Hubble rate H .
- This proportionality arises from the scaling of stress-tensor correlators in an expanding Universe.
- No heuristic geometric-mean arguments are required.
- The result follows directly from the operator dimension, the FRW background, and the Born–Markov structure.

The next section projects the Lindblad equation onto the off-diagonal components of the global density matrix and derives the effective RDCC relaxation equation.

4 Effective Lindblad and Boltzmann Equation for Off-Diagonal Elements

The Lindblad equation (15) governs the full bipartite density matrix in the interaction picture. To connect with the RDCC relaxation dynamics, we now project this equation onto the off-diagonal components of the global density matrix in the sectoral basis.

4.1 Sectoral Basis and Coherence Components

Let $\{|+\rangle\}$ and $\{|-\rangle\}$ denote orthonormal bases of the two sectors. The global density matrix can be decomposed as

$$\rho_{\text{global}} = \begin{pmatrix} \rho_{++} & \rho_{+-} \\ \rho_{-+} & \rho_{--} \end{pmatrix}, \quad (20)$$

where the off-diagonal blocks ρ_{+-} and ρ_{-+} encode inter-sector coherence.

The relaxation dynamics of RDCC are governed precisely by the decay of these off-diagonal components.

4.2 Projection of the Lindblad Equation

The Lindblad equation (15) contains terms of the form

$$L_A \rho L_A^\dagger, \quad L_A^\dagger L_A \rho, \quad \rho L_A^\dagger L_A.$$

Because the Lindblad operators L_A are constructed from stress-tensor components of $\mathcal{H}_+$ and $\mathcal{H}_-$, they act nontrivially on both sectors and mix the off-diagonal blocks.

Projecting Eq. (15) onto the $(+-)$ block yields

$$\frac{d}{dt} \rho_{+-}(t) = -\Gamma(t) \rho_{+-}(t) + i\Omega(t) \rho_{+-}(t), \quad (21)$$

where:

- $\Gamma(t)$ is the decoherence (relaxation) rate,
- $\Omega(t)$ is a phase-shift term arising from the anti-Hermitian part of the kernel.

The phase term $\Omega(t)$ does not affect the magnitude of ρ_{+-} and plays no role in the RDCC relaxation parameter α . We therefore focus on the real, dissipative part.

4.3 Explicit Form of the Relaxation Rate

From Eq. (15), the relaxation rate is

$$\Gamma(t) \sim \frac{1}{M^4 H(t)} \langle TT \rangle_{\text{phys}}^2(t). \quad (22)$$

Using the scaling $\langle TT \rangle \sim H^4 M_{\text{Pl}}^4$, we recover the proportionality

$$\Gamma(t) \propto H(t), \quad (23)$$

as derived in Sec. 3.

4.4 Boltzmann-Type Formulation

Equation (21) can be rewritten as a Boltzmann-type equation for the coherence amplitude $\mathcal{C}(t) = \text{Tr}(\rho_{+-})$:

$$\frac{d}{dt} \mathcal{C}(t) = -\Gamma(t) \mathcal{C}(t). \quad (24)$$

This is the microscopic origin of the RDCC relaxation equation $\dot{\alpha} = \Gamma\alpha$.

4.5 Solution and Physical Interpretation

The solution of Eq. (24) is

$$\mathcal{C}(t) = \mathcal{C}(t_0) \exp \left[- \int_{t_0}^t \Gamma(t') dt' \right]. \quad (25)$$

Because $\Gamma(t) \propto H(t)$, the decay of inter-sector coherence is tied directly to the cosmological expansion rate. This provides a structural explanation for why the RDCC relaxation parameter α evolves multiplicatively with $\Gamma \propto H$.

4.6 Summary

Projecting the Lindblad equation onto the off-diagonal components of the global density matrix yields:

- a first-order decay equation for inter-sector coherence,
- a relaxation rate proportional to the Hubble rate,
- and a Boltzmann-type formulation consistent with the RDCC relaxation equation.

The next section evaluates the relaxation kernel explicitly and connects the microscopic derivation to the phenomenological RDCC parameter α .

5 Microscopic Evaluation of the Relaxation Kernel

We now evaluate the relaxation kernel appearing in the Lindblad equation and show explicitly how the proportionality $\Gamma \propto H$ emerges from the scaling of stress-tensor correlators in an expanding Universe. This section provides the microscopic foundation for the RDCC relaxation equation $\dot{\alpha} = \Gamma\alpha$.

5.1 Structure of the Kernel

The Born–Markov kernel entering Eq. (15) is

$$K(t) = \int_0^{H^{-1}} d\tau \int d^3x a(t)^3 \int d^3y a(t-\tau)^3 \langle T_{\mu\nu}^{(+)}(t, \vec{x}) T_{\rho\sigma}^{(+)}(t-\tau, \vec{y}) \rangle \langle T_{\mu\nu}^{(-)}(t, \vec{x}) T_{\rho\sigma}^{(-)}(t-\tau, \vec{y}) \rangle. \quad (26)$$

The upper limit of integration is set by the Hubble memory time, as established in Sec. 3. The kernel is dominated by modes with physical frequencies $\omega_{\text{phys}} \sim H$, since faster modes dephase and slower modes contribute negligibly to the commutator.

5.2 Scaling of Stress-Tensor Correlators

For a relativistic fluid in an FRW background, the stress tensor scales as

$$T_{\mu\nu} \sim \rho \sim 3M_{\text{Pl}}^2 H^2. \quad (27)$$

The equal-time correlator therefore scales as

$$\langle T_{\mu\nu}(t) T_{\rho\sigma}(t) \rangle \sim H^4 M_{\text{Pl}}^4. \quad (28)$$

The time-separated correlator inherits an exponential decay on the Hubble timescale,

$$\langle T(t) T(t-\tau) \rangle \sim H^4 M_{\text{Pl}}^4 e^{-H\tau}, \quad 0 \leq \tau \lesssim H^{-1}. \quad (29)$$

5.3 Evaluation of the Kernel

Inserting this scaling into Eq. (26) yields

$$K(t) \sim \int_0^{H^{-1}} d\tau a(t)^3 a(t-\tau)^3 (H^4 M_{\text{Pl}}^4)^2 e^{-2H\tau}. \quad (30)$$

Over a Hubble time, the scale factor changes only by order unity, so $a(t)^3 a(t-\tau)^3 \sim a(t)^6$ up to $\mathcal{O}(1)$ corrections. Thus

$$K(t) \sim a(t)^6 H^8 M_{\text{Pl}}^8 \int_0^{H^{-1}} d\tau e^{-2H\tau} \sim a(t)^6 H^7 M_{\text{Pl}}^8. \quad (31)$$

5.4 Relaxation Rate

The relaxation rate is obtained by inserting Eq. (31) into the Lindblad prefactor,

$$\Gamma(t) \sim \frac{1}{M^4} K(t) \sim \frac{a(t)^6}{M^4} H^7 M_{\text{Pl}}^8. \quad (32)$$

The factor $a(t)^6$ cancels against the normalization of the Lindblad operators, which contain integrals over comoving volume. After normalization, the physical relaxation rate becomes

$$\Gamma(t) \sim \frac{M_{\text{Pl}}^8}{M^4} H(t). \quad (33)$$

Up to a dimensionless coefficient and the ratio M_{Pl}^8/M^4 , the relaxation rate is *proportional to the Hubble rate*. This is the microscopic origin of the RDCC scaling relation.

6 Discussion and Embedding into the RDCC Framework

The microscopic derivation presented in this Companion establishes a direct link between the dimension-six inter-sector operator and the relaxation dynamics central to the Relaxation-Driven Cyclic Cosmology (RDCC) framework. The key result is the proportionality

$$\Gamma \propto H, \quad (34)$$

which emerges naturally from the scaling of stress-tensor correlators and the Hubble memory time of the FRW background.

This result provides the structural foundation for the multiplicative relaxation equation $\dot{\alpha} = \Gamma\alpha$ used throughout the RDCC series. It also clarifies why the relaxation parameter $\alpha(a)$ freezes out at early times: once the Hubble rate drops below the characteristic scale set by the correlators, the relaxation rate becomes too small to drive further evolution.

6.1 Consistency with Earlier Companions

The microscopic picture developed here is consistent with the dynamical results of Companions III–VII:

- The freeze-out of $\alpha(a)$ (Companion VI) follows from the decay of $\Gamma \propto H$ as the Universe expands.
- The log-periodic tensor modulation (Companions II–V) arises from interference between the two sectors, whose coherence is governed by the same relaxation rate.
- The stability of the RDCC prediction vector (Companion VII) reflects the fact that the relaxation dynamics are universal and insensitive to microscopic details.

Thus the microscopic derivation of Γ presented here is not an isolated result but a structural completion of the RDCC framework.

6.2 Interpretation of the Dimension-Six Operator

The operator $\mathcal{L}_{\text{int}} = M^{-2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}$ plays a dual role:

1. It governs the decay of inter-sector coherence through the Lindblad kernel.
2. It induces the interference responsible for tensor-sector modulation.

Both effects arise from the same correlator structure. This unifies the relaxation dynamics and the phenomenology of primordial tensor modes under a single microscopic mechanism.

6.3 Emergence of Macroscopic Structure

The proportionality $\Gamma \propto H$ implies that decoherence is driven by the cosmological expansion itself. This provides a natural explanation for why classical spacetime, causal structure, and thermodynamic arrows emerge only after sectoral projection: the expansion continuously suppresses inter-sector coherence, making the global state effectively classical on Hubble timescales.

6.4 Limitations and Future Directions

The derivation presented here relies on:

- the Born approximation (weak coupling),
- the Markov approximation (Hubble-scale memory time),
- and the scaling of stress-tensor correlators for relativistic species.

A more complete treatment would include:

- non-Markovian corrections,
- mode-dependent relaxation rates,
- and a full information-theoretic analysis of inter-sector entanglement.

These extensions are natural directions for future Companions.

A particularly important limitation of the present derivation is the implicit assumption that the Hubble rate provides a finite and monotonic memory scale. Exactly at the CPT-symmetric bounce, however, the Hubble parameter vanishes, $H(t_b) = 0$, and the Markov approximation formally breaks down. The stress-tensor correlator retains a finite temporal support around the bounce, and the replacement of the memory kernel by a delta function is no longer valid. This motivates the non-Markovian refinement developed in Sec. 7, where a compact memory window of width Δt_{mem} regularizes the kernel and yields a finite, symmetric correction to the initial relaxation dynamics. This correction does not alter the large-scale scaling $\Gamma \propto H$ but provides a controlled mechanism for generating the primordial coherence structure of the mirror sector.

7 Non-Markovian Memory Window at the Bounce

The microscopic derivation presented in this Companion relies on the Born–Markov approximation, with the Hubble time H^{-1} acting as the natural memory scale of the FRW background. This approximation is valid throughout the expanding and contracting phases, where the Hubble rate provides a monotonic coarse-graining window. However, exactly at the CPT-symmetric bounce, where $H(t_b) = 0$, the Markov assumption breaks down for an ultrashort interval. In this section we quantify this breakdown and derive the leading non-Markovian correction to the inter-sector relaxation kernel.

7.1 Breakdown of the Markov Approximation at $H = 0$

The Markov approximation assumes that the stress-tensor correlator $C(t, \tau)$ decays on a timescale $\tau \sim H^{-1}(t)$. At the bounce, the physical Hubble rate vanishes,

$$H(t_b) = 0, \quad (35)$$

and the memory time formally diverges. The correlator therefore retains a finite temporal support around the bounce, and the standard replacement

$$\int_0^\infty d\tau C(t, \tau) \longrightarrow C(t, 0) H^{-1}(t)$$

is no longer valid. Instead, the kernel acquires a compact, non-Markovian memory window of width Δt_{mem} determined by the curvature of the scale factor.

To leading order, the scale factor near the bounce may be expanded as

$$a(t) = a_b + \frac{1}{2} \ddot{a}_b (t - t_b)^2 + \mathcal{O}((t - t_b)^3), \quad (36)$$

with $\ddot{a}_b > 0$ for a non-singular bounce. The physical frequency of stress-tensor modes therefore varies quadratically in $(t - t_b)$, and the correlator decays on a timescale

$$\Delta t_{\text{mem}} \sim (\ddot{a}_b)^{-1/2}, \quad (37)$$

which replaces the divergent Hubble time.

7.2 Minimal Non-Markovian Kernel

To capture the leading correction, we introduce a Gaussian memory kernel

$$K_{\text{NM}}(t, \tau) = \exp \left[-\frac{\tau^2}{2 \Delta t_{\text{mem}}^2} \right], \quad (38)$$

which reduces to the Markovian delta kernel in the limit $\Delta t_{\text{mem}} \rightarrow 0$. The Born expansion of the interaction-picture evolution then yields

$$\frac{d}{dt} \rho(t) = -\frac{1}{M^4} \int_0^\infty d\tau K_{\text{NM}}(t, \tau) [T(t), [T(t - \tau), \rho(t)]], \quad (39)$$

where T denotes the stress-tensor operator of the interacting sectors.

Equation (39) is completely positive and reduces to the Lindblad form away from the bounce. The non-Markovian correction is confined to a narrow interval $|t - t_b| \lesssim \Delta t_{\text{mem}}$.

7.3 Correction to the Relaxation Rate

Projecting Eq. (39) onto the off-diagonal block ρ_{+-} yields a modified relaxation equation

$$\frac{d}{dt} \rho_{+-}(t) = -\Gamma_{\text{eff}}(t) \rho_{+-}(t), \quad (40)$$

with effective rate

$$\Gamma_{\text{eff}}(t) = \Gamma_{\text{Markov}}(t) \left[1 + \varepsilon_{\text{NM}} \exp \left(-\frac{(t - t_b)^2}{2 \Delta t_{\text{mem}}^2} \right) \right], \quad (41)$$

where ε_{NM} is a dimensionless coefficient determined by the curvature of the scale factor and the stress-tensor correlator.

The correction is localized at the bounce and integrates to a finite shift in the initial coherence amplitude. Physically, this corresponds to a brief “memory echo” in which the two sectors exchange information more efficiently than in the Markovian regime.

7.4 Physical Interpretation

The non-Markovian window has three key implications:

1. It provides a controlled mechanism for generating primordial fluctuations in the mirror sector without imposing them as boundary data.
2. It regularizes the Markovian divergence at $H = 0$ and yields a finite, well-defined initial relaxation kernel.
3. It preserves the global CPT symmetry: the memory window is symmetric under $t - t_b \mapsto -(t - t_b)$.

Away from the bounce, the standard RDCC scaling $\Gamma \propto H$ is recovered exactly. The non-Markovian correction therefore acts as a localized, structural refinement of the microscopic relaxation dynamics.

8 Outlook

The microscopic derivation of the RDCC relaxation rate presented in this Companion opens several avenues for further development:

Information-theoretic formulation. A rigorous treatment of inter-sector entanglement entropy and its evolution under the dimension-six operator would clarify the connection between decoherence, causal structure, and emergent spacetime.

Tensor-sector phenomenology. The same correlators that govern relaxation also shape the tensor modulation pattern. Future gravitational-wave observations may therefore probe the microscopic coherence structure of the RDCC state.

Emergent geometry. The relaxation dynamics suggest that aspects of large-scale geometry may be understood in terms of correlation decay rather than fundamental spacetime primitives. This perspective aligns with the global CPT-symmetric structure of RDCC.

Beyond the Markov approximation. Including non-Markovian effects may reveal additional structure in the relaxation kernel, particularly near the bounce where the Hubble rate changes rapidly.

Numerical implementation. A full Boltzmann-code implementation of the RDCC relaxation dynamics would allow for global parameter scans and quantitative comparison with cosmological data.

In summary, the microscopic derivation of $\Gamma \propto H$ presented here provides the missing link between the dimension-six operator and the phenomenological relaxation dynamics used throughout the RDCC framework. It unifies decoherence, expansion, and inter-sector coupling under a single structural mechanism and sets the stage for future developments in the information-theoretic foundations of RDCC.

Appendix A: Projective Compactification of the CPT Fixed Point

The global RDCC state lives on the bipartite Hilbert space $\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$, with the two sectors related by global CPT symmetry. At the cosmological bounce, the relaxation parameter satisfies $\alpha(t_b) \rightarrow 0$, the two sectors become maximally entangled, and the coarse-grained entropy of the global state reaches its minimum. This hypersurface is the structural fixed point of the CPT symmetry.

To make this identification geometrically explicit, we introduce a projective compactification of the time coordinate using the real projective line $\mathbb{RP}^1$, the one-point compactification of $\mathbb{R}$ in which $+\infty$ and $-\infty$ are identified. Points on $\mathbb{RP}^1$ are equivalence classes $[x : y]$ with $[x : y] = [\lambda x : \lambda y]$ for $\lambda \neq 0$, and the ordinary reals embed via $t \mapsto [t : 1]$. The point at infinity corresponds to $[1 : 0]$.

To describe the neighbourhood of the bounce, we define a projective deviation coordinate

$$\tau = \frac{1}{a - a_b},$$

with $a_b = 1$ the minimal scale factor. As $a \rightarrow 1^+$ (the $+$ sector), one has $\tau \rightarrow +\infty$; as $a \rightarrow 1^-$ (the $-$ sector), one has $\tau \rightarrow -\infty$. Both limits map to the same projective point:

$$\lim_{\tau \rightarrow \pm\infty} [1 : 1/\tau] = [1 : 0].$$

Thus the bounce corresponds to the projective point where $+\infty$ and $-\infty$ are identified. In these coordinates, the global CPT operator acts as $\tau \mapsto -\tau$, enforcing

$$\rho_+(\tau) = \Theta \rho_-(-\tau) \Theta^\dagger.$$

At the fixed point $\tau = 0$, the two reduced density matrices coincide, and the entanglement entropy is maximal, consistent with $\alpha(t_b) = 0$.

The relaxation dynamics acquire a geometric interpretation: the multiplicative evolution

$$\dot{\alpha} = \Gamma \alpha, \quad \Gamma \propto H,$$

describes the flow of the coherence parameter away from the projective fixed point. As the Universe moves along the compactified line, the distance from $\tau = 0$ increases, and the two sectors decohere at a rate set by the Hubble expansion.

This compactification regularizes the bounce as a removable singularity in the global state: the two time directions are glued into a single smooth point on $\mathbb{RP}^1$, as required by CPT symmetry. No new fields or interactions are introduced; the construction provides a natural topological language for the bipartite structure and the role of the bounce as CPT fixed point.

Finally, the projective picture connects directly to the coherence-boundary interpretation of the invariant speed c . The characteristic coherence length

$$\ell_{\text{coh}} \sim \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}}$$

emerges naturally as the scale controlling how rapidly correlations can propagate across the projective fixed point before relaxation enforces sectoral projection.

This geometric perspective complements the dynamical and microscopic derivations presented in the main text and clarifies the structural unity of the RDCC framework.